TRANSMISSION—5R55N—SHIFT ADAPTIVE ELECTRONIC PRESSURE CONTROL STRATEGY—SERVICE TIP	Article No. 99-23-1

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This TSB article is being made available to use as an aid in understanding and diagnosing pressure concerns on vehicles equipped with Shift Adaptive Electronic Pressure Control Strategy.

ACTION

Refer to the following text when servicing pressure concerns on vehicles equipped with Shift Adaptive Electronic Pressure Control Strategy.

SERVICE PROCEDURE

Follow normal diagnostic procedures as listed in the 2000 LS Workshop Manual. Make sure the electronic engine control inputs that determine shift adaptive electronic pressure control strategy, which are listed below, are operating properly. Another check is to determine if the vehicle battery has been recently disconnected.

SHIFT ADAPTIVE ELECTRONIC PRESSURE CONTROL STRATEGY DESCRIPTION

The 2000 Lincoln LS and Jaguar S-Type use an adaptive electronic pressure control strategy to allow better control over transmission shift events. This strategy has been implemented for the following reasons:

- Maintain high mileage durability
- Reduce unit variation from transmission to transmission
- Provide consistent shift feel over the life of the vehicle

Based on input signals, the strategy calculates an actual shift time and compares it to an ideal shift time. Once the vehicle is up to operating temperature, incremental pressure changes are made to minimize the difference between actual and ideal shift times.

This strategy uses the following inputs for adaptive electronic pressure control:

- Mass air flow
- Engine speed
- Turbine speed

Adaptive electronic pressure control strategy modifies shifting pressures when certain operating conditions are met. Beyond operating temperature, a minimum throttle level and a minimum change in throttle must also be met.

When these conditions are met, the following shift events will adapt:

- Upshifts
- Closed throttle, manual downshifts
- Engagements

It is normal for a brand new vehicle/transmission to have abrupt shift events. Once the vehicle is at operating temperature, it may take several shifts at the same operating conditions for the transmission to properly adapt. The incremental pressure change information gets stored in Keep Alive Memory (KAM). If for any reason the vehicle loses power to KAM, the transmission will again shift in its pre-adapted level.

The following are possible reasons that KAM may be cleared:

- Vehicle battery is disconnected, or battery is drained
- An updated calibration is programmed into the Powertrain Control Module (PCM)
- KAM power is lost to the PCM while the battery is still connected
- KAM may be cleared while using diagnostic equipment

The adaptive learning process may take anywhere from 80 to 320 km (50 to 200 miles) of stop-and-go driving to fully update KAM, depending on driving style. The more varied the speed and torque during vehicle operation, the more complete the adaptive learning process will be.

OTHER APPLICABLE ARTICLES: NONE
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